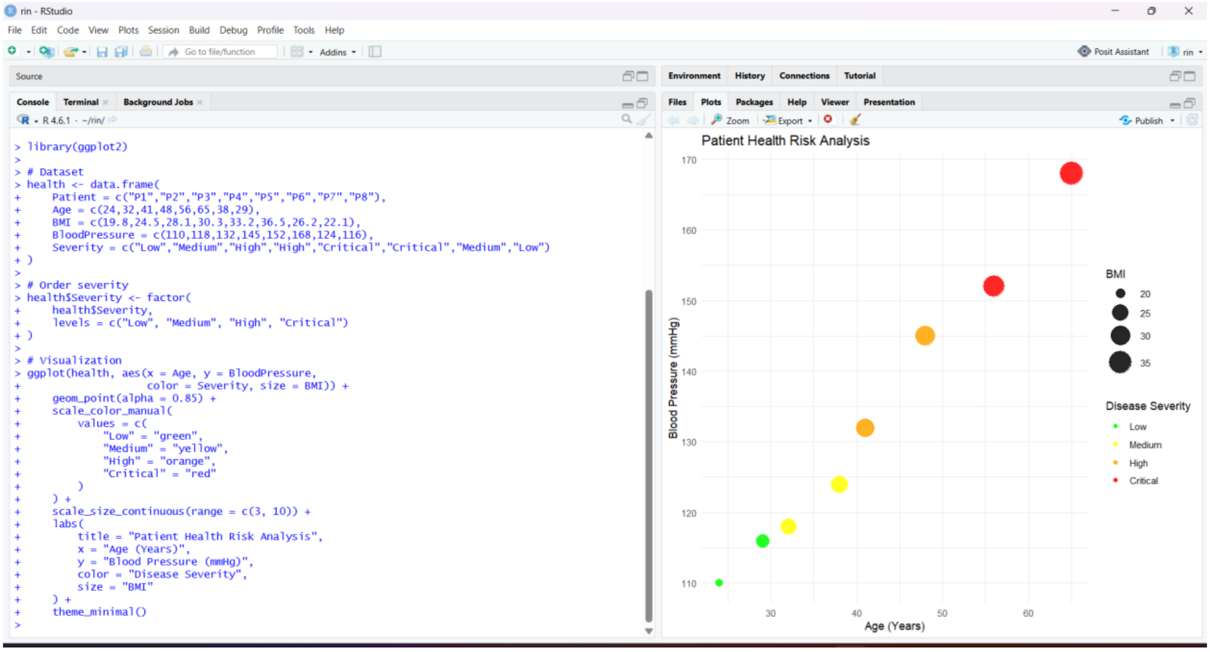
Lab Experiments of Data Handling and Visualization

Q1.

Code:

```
library(ggplot2)
# Dataset
health <- data.frame(
  Patient = c("P1","P2","P3","P4","P5","P6","P7","P8"),
  Age = c(24,32,41,48,56,65,38,29),
  BMI = c(19.8,24.5,28.1,30.3,33.2,36.5,26.2,22.1),
  BloodPressure = c(110,118,132,145,152,168,124,116),
  Severity = c("Low","Medium","High","High","Critical","Critical","Medium","Low")
)
# Order severity
health$Severity <- factor(
  health$Severity,
  levels = c("Low", "Medium", "High", "Critical")
)
# Visualization
ggplot(health, aes(x = Age, y = BloodPressure,
                   color = Severity, size = BMI)) +
  geom_point(alpha = 0.85) +
  scale_color_manual(
    values = c(
      "Low" = "green",
      "Medium" = "yellow",
      "High" = "orange",
      "Critical" = "red"
    )
  ) +
```

```
scale_size_continuous(range = c(3, 10)) +
labs(
  title = "Patient Health Risk Analysis",
  x = "Age (Years)",
  y = "Blood Pressure (mmHg)",
  color = "Disease Severity",
  size = "BMI"
) +
theme_minimal()
```



Q2.

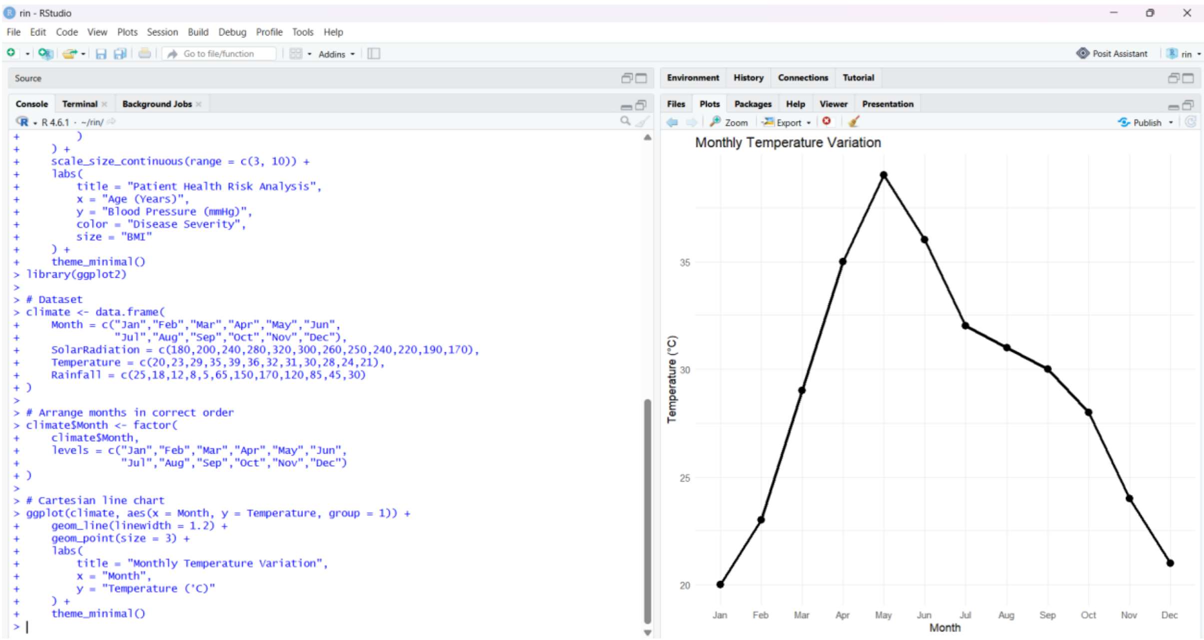
Code:

Temperature:

```
library(ggplot2)

# Dataset
climate <- data.frame(
  Month = c("Jan","Feb","Mar","Apr","May","Jun",
          "Jul","Aug","Sep","Oct","Nov","Dec"),
  SolarRadiation = c(180,200,240,280,320,300,260,250,240,220,190,170),
  Temperature = c(20,23,29,35,39,36,32,31,30,28,24,21),
```

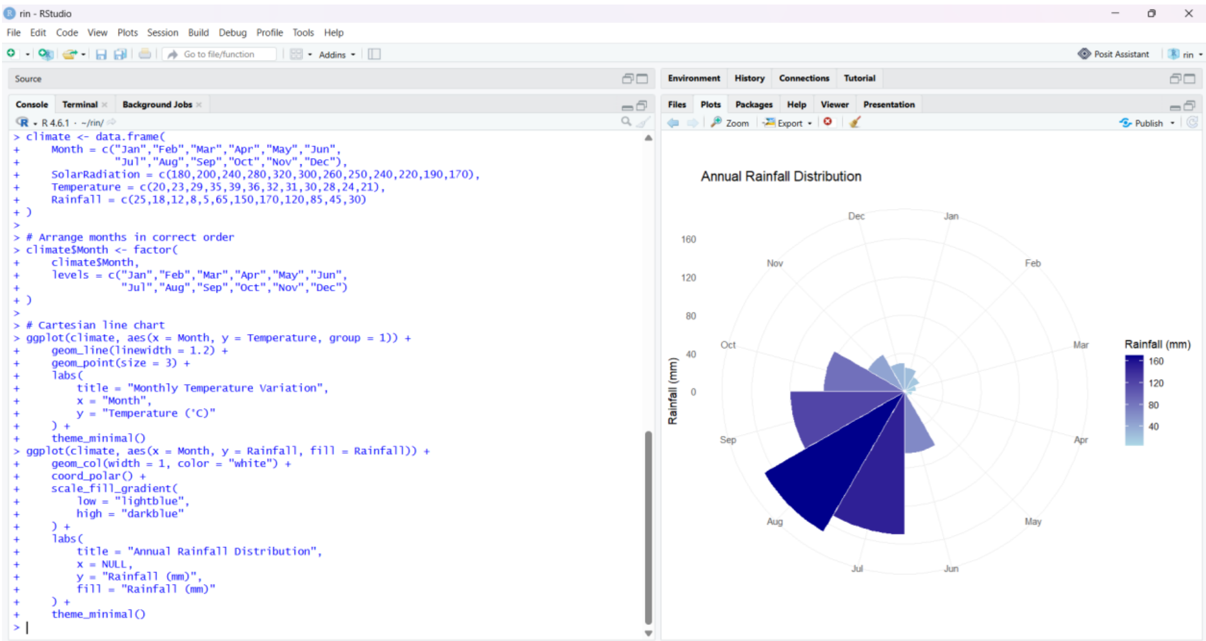
```
  Rainfall = c(25,18,12,8,5,65,150,170,120,85,45,30)
)
# Arrange months in correct order
climate$Month <- factor(
  climate$Month,
  levels = c("Jan","Feb","Mar","Apr","May","Jun",
             "Jul","Aug","Sep","Oct","Nov","Dec")
)
# Cartesian line chart
ggplot(climate, aes(x = Month, y = Temperature, group = 1)) +
  geom_line(linewidth = 1.2) +
  geom_point(size = 3) +
  labs(
    title = "Monthly Temperature Variation",
    x = "Month",
    y = "Temperature (°C)"
  ) +
  theme_minimal()
```



Rain Fall:

Code:

```
ggplot(climate, aes(x = Month, y = Rainfall, fill = Rainfall)) +
  geom_col(width = 1, color = "white") +
  coord_polar() +
  scale_fill_gradient(
    low = "lightblue",
    high = "darkblue"
  ) +
  labs(
    title = "Annual Rainfall Distribution",
    x = NULL,
    y = "Rainfall (mm)",
    fill = "Rainfall (mm)"
  ) +
  theme_minimal()
```



Q3.

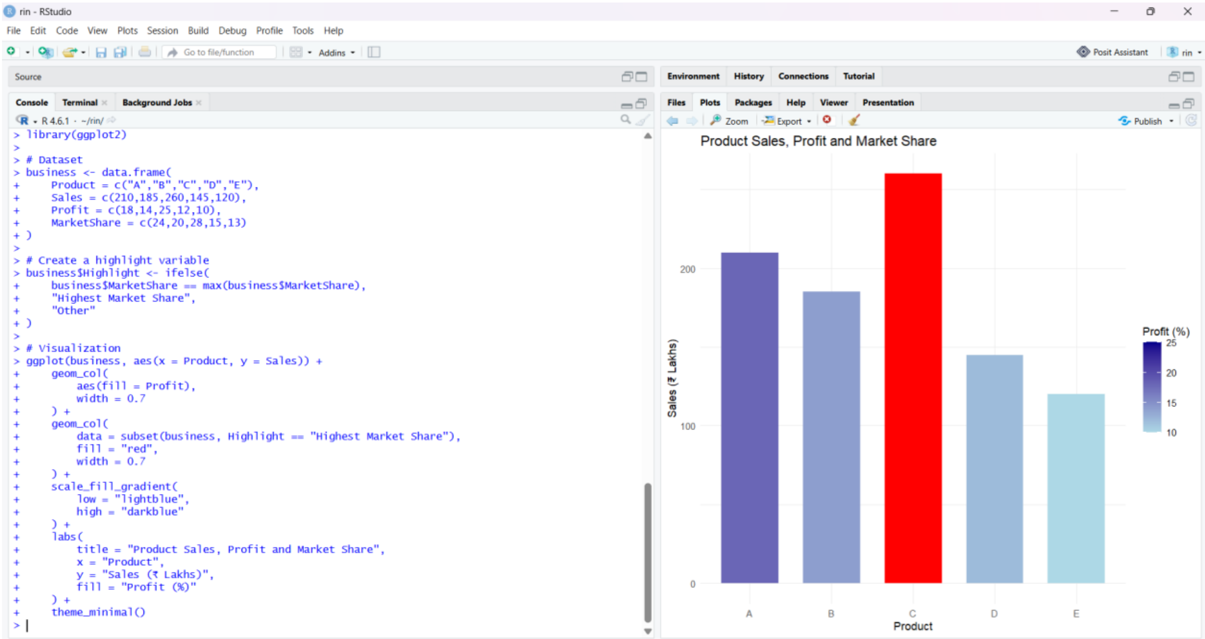
Code:

library(ggplot2)

Dataset

```
business <- data.frame(
  Product = c("A","B","C","D","E"),
  Sales = c(210,185,260,145,120),
  Profit = c(18,14,25,12,10),
  MarketShare = c(24,20,28,15,13)
)
# Create a highlight variable
business$Highlight <- ifelse(
  business$MarketShare == max(business$MarketShare),
  "Highest Market Share",
  "Other"
)
# Visualization
ggplot(business, aes(x = Product, y = Sales)) +
  geom_col(
    aes(fill = Profit),
    width = 0.7
  ) +
  geom_col(
    data = subset(business, Highlight == "Highest Market Share"),
    fill = "red",
    width = 0.7
  ) +
  scale_fill_gradient(
    low = "lightblue",
    high = "darkblue"
  ) +
  labs(
    title = "Product Sales, Profit and Market Share",
```

```
    x = "Product",
    y = "Sales (₹ Lakhs)",
    fill = "Profit (%)"
  ) +
  theme_minimal()
```

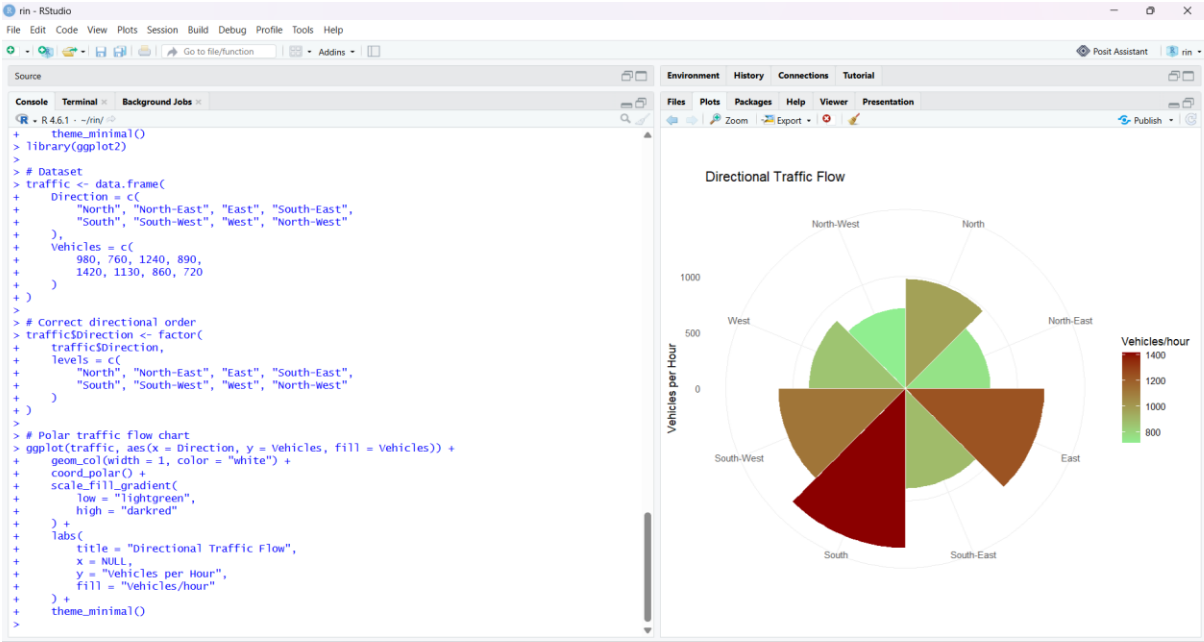


Q4.

Code:

```
library(ggplot2)
# Dataset
traffic <- data.frame(
  Direction = c(
    "North", "North-East", "East", "South-East",
    "South", "South-West", "West", "North-West"
  ),
  Vehicles = c(
    980, 760, 1240, 890,
    1420, 1130, 860, 720
  )
)
```

```
# Correct directional order
traffic$Direction <- factor(
  traffic$Direction,
  levels = c(
    "North", "North-East", "East", "South-East",
    "South", "South-West", "West", "North-West"
  )
)
# Polar traffic flow chart
ggplot(traffic, aes(x = Direction, y = Vehicles, fill = Vehicles)) +
  geom_col(width = 1, color = "white") +
  coord_polar() +
  scale_fill_gradient(
    low = "lightgreen",
    high = "darkred"
  ) +
  labs(
    title = "Directional Traffic Flow",
    x = NULL,
    y = "Vehicles per Hour",
    fill = "Vehicles/hour"
  ) +
  theme_minimal()
```



Q5.

Code:

```
library(ggplot2)

library(patchwork)

# Dataset

environment <- data.frame(

  City = c("A","B","C","D","E","F"),

  AQI = c(48,82,135,165,72,95),

  GreenCover = c(45,32,21,18,38,30),

  CO2 = c(1.8,3.5,5.6,7.4,2.9,4.2),

  Population = c(9,15,22,28,12,18)

)


# Create AQI categories

environment$AQI_Category <- cut(

  environment$AQI,

  breaks = c(-Inf, 50, 100, 150, 200, Inf),

  labels = c(

    "Good",
```

```
    "Moderate",
    "Unhealthy",
    "Very Unhealthy",
    "Hazardous"
  )
)
# Plot 1: AQI by City
p1 <- ggplot(
  environment,
  aes(x = City, y = AQI, fill = AQI_Category)
) +
  geom_col(width = 0.7) +
  scale_fill_manual(
    values = c(
      "Good" = "green",
      "Moderate" = "yellow",
      "Unhealthy" = "orange",
      "Very Unhealthy" = "red",
      "Hazardous" = "darkred"
    )
  ) +
  labs(
    title = "Air Quality Index by City",
    x = "City",
    y = "AQI",
    fill = "AQI Category"
  ) +
  theme_minimal()
# Plot 2: Green Cover vs AQI
```

```
p2 <- ggplot(
  environment,
  aes(
    x = GreenCover,
    y = AQI,
    size = Population,
    color = AQI_Category
  )
) +
  geom_point(alpha = 0.8) +
  scale_size_continuous(range = c(4, 12)) +
  scale_color_manual(
    values = c(
      "Good" = "green",
      "Moderate" = "yellow",
      "Unhealthy" = "orange",
      "Very Unhealthy" = "red",
      "Hazardous" = "darkred"
    )
  ) +
  labs(
    title = "Green Cover vs Air Quality",
    x = "Green Cover (%)",
    y = "AQI",
    size = "Population (Lakhs)",
    color = "AQI Category"
  ) +
  theme_minimal()
# Plot 3: CO2 Emission
```

```
p3 <- ggplot(
  environment,
  aes(x = City, y = CO2, fill = CO2)
) +
  geom_col(width = 0.7) +
  scale_fill_gradient(
    low = "lightgreen",
    high = "darkred"
  ) +
  labs(
    title = "CO2 Emission by City",
    x = "City",
    y = "CO2 Emission",
    fill = "CO2"
  ) +
  theme_minimal()
# Plot 4: Population
p4 <- ggplot(
  environment,
  aes(x = City, y = Population, fill = Population)
) +
  geom_col(width = 0.7) +
  scale_fill_gradient(
    low = "lightblue",
    high = "darkblue"
  ) +
  labs(
    title = "Population by City",
    x = "City",
```

```
    y = "Population (Lakhs)",
    fill = "Population"
  ) +
  theme_minimal()
# Combine into dashboard
(p1 | p2) /
(p3 | p4)
```